ANULUM / SCIENTIFIC DISCOVERY AND EXPERIMENTATION

The discovery cycle being built

Protocols, domain execution, replay and model ownership · work in progress · 13 October 2026

HUMAN DECISION POINTS
Protocol freeze + specified computation scope + outward release / publication

01
Question and protocol
Scientific question and hypothesis
Domain oracle and validity scope
Metric, threshold, seeds and budget

Artifact: protocol + source references

02
Review and freeze
Domain owner + second reviewer
Verdicts on exact protocol bytes
Human approval of the frozen scope

Artifact: digest + approval record

03
Request and acceptance
Versioned request to the domain
Domain may accept or decline
Inputs, compute and hardware limits

Artifact: accepted / declined request

04
Domain experiment
Run within the authorised scope
Retain attempt and input identity
Initialisation, errors and raw output

Artifact: attempt + raw records

Retained attempt records enter the evidence and verdict path

05
Domain checks
Compare to the named reference
Apply validity and failure criteria
Keep negative and null outcomes

Artifact: domain verification report

06
Scientific verdict
Domain owns the scientific decision
State pass, failure or uncertainty
Report where the result is valid

Artifact: signed / attributed verdict

07
Custody and replay
Check schema, hashes and provenance
Replay the specified derivation
Loop admission follows domain verdict

Artifact: admission + replay record

08
Record and next model
Append the outcome and evidence
Propose a domain-owned model update
Use the record to form the next question

Artifact: lineage + update proposal

Next question comes from the retained outcome, including a negative result

Discovery group
Owns the loop and the common yardstick.
Owns the benchmark / report format.
Can refuse a record for custody or replay failure.
Does not change the domain’s scientific verdict.

Domain group
Owns models, physics, oracles and calibration.
Owns measurement and hardware authority.
Executes or declines the experiment request.
Accepts its own model revisions and validity claims.

Human authority
Sets rules, budget and assigned authority.
Approves freeze and named computation scope.
Decides releases and publications.
A protocol approval is not a physical-action grant.

EVIDENCE PACKAGE — WHAT THE NEXT SESSION OR REPLICATOR RECEIVES

Protocol digest · input and attempt identifiers · seeds and budgets · domain validity criteria
Raw retained output · simulator / instrument status · calibration and version context · scientific verdict
Deterministic derivation · content hashes · replay result · custody and reviewer attribution
Outcome and model-update lineage · failure history · cost and energy fields as measurement is integrated

FIXED GRID-DETECTOR EXPERIMENT

Current frame: v0.15, review and harness work.
Fixed aggregation; valid initialisation required.
Held-out simulated systems; retained failure outcomes.
In progress: protocol, adapter and replay-harness work.

NEURAL-DYNAMICS EXPERIMENT

Current source frame: v0.5, review and preparation.
AdEx surrogate with a scoped validity boundary.
Preregistered protocol, independent checks and replay.
Domain choices remain with the neuromorphic group.

Sources: discovery group boundaries, current protocol frames and the research investment plan; source digests embedded in the SVG metadata.

Technical interfaces, record formats and the full cycle are in progress. Scientific execution follows the authorised freeze.